Chapter 18 · Semester 2

ALTERNATING CURRENT CIRCUITS

Notes by Cikgu Suhaili

9 worked examples 3 simulations Exercises with answers

18.1 Alternating current through a resistor

An **alternating current** (a.c.) changes its magnitude and direction periodically. The output of an a.c. generator is sinusoidal:

$$V = V_0 \sin \omega t \qquad I = I_0 \sin \omega t \qquad \omega = 2\pi f = \frac{2\pi}{T}$$

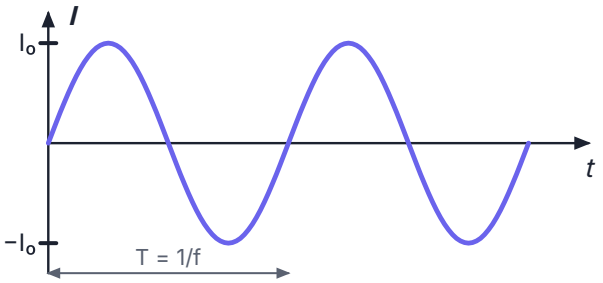


Figure 18.1: a sinusoidal alternating current with peak value I_o and period T.

- V, I: instantaneous values; V_0, I_0: peak values; ω: angular frequency (rad s^{-1}); f: frequency (Hz).

Phasors

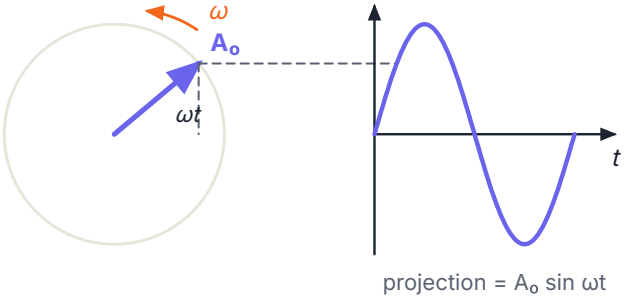


Figure 18.2: a phasor of length A_o rotating anticlockwise at ω. Its vertical projection is A_o sin ωt.

A **phasor** is a vector that rotates anticlockwise at the angular frequency ω. Its length is the peak value and its projection on the vertical axis gives the instantaneous value. A **phasor diagram** shows the phase difference between voltage and current.

Root mean square (r.m.s.) value

r.m.s. current, I_{rms}

the **value of the steady direct current that produces the same average power** in a resistor as the alternating current. (The r.m.s. voltage is defined in the same way.)

Power in a resistor at any instant: $P = I^2R$. The average power is $\langle I^2 \rangle R$.

A steady current I_{rms} gives the same power if $I_{rms}^2 R = \langle I^2 \rangle R$, so $I_{rms} = \sqrt{\langle I^2 \rangle}$: the **square root of the mean of the square** of the current.

For $I = I_0 \sin \omega t$: $I^2 = I_0^2 \sin^2 \omega t = \frac{1}{2} I_0^2 (1 - \cos 2\omega t)$. Over a whole cycle the mean of $\cos 2\omega t$ is zero, so $\langle I^2 \rangle = \frac{1}{2} I_0^2$.

$$I_{rms} = \frac{I_0}{\sqrt{2}} \qquad V_{rms} = \frac{V_0}{\sqrt{2}}$$

sinusoidal a.c. only. Mains 240 V means $V_{rms} = 240$ V, so $V_0 = 339$ V.

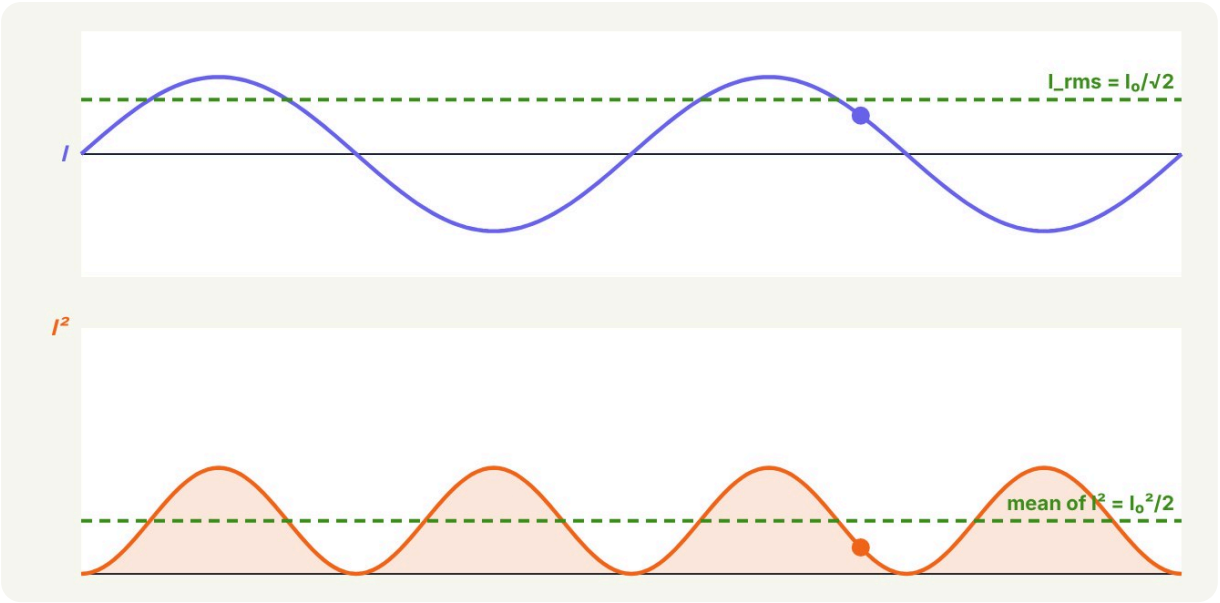
Teacher's note

Ammeters and voltmeters for a.c. read r.m.s. values. Unless a question says "peak", assume a quoted a.c. voltage or current is r.m.s.

Simulation

Where the r.m.s. value comes from

Watch the current, its square and the mean of the square. The steady current that gives the same heating is the square root of that mean: $I_o/\sqrt{2}$ for a sine wave.



I_o	mean of I^2	$I_{rms} = \sqrt{(\text{mean of } I^2)}$	mean of I itself
4.0 A	**8.00 A²**	**2.83 A**	**0 (no use for heating)**

This is a still picture. Open PhysiQ › Notes › Semester 2 › Chapter 18 to use the simulation.

Current in a pure resistor

Supply voltage across the resistor: $V = V_0 \sin \omega t$.

At every instant $I = \dfrac{V}{R} = \dfrac{V_0}{R} \sin \omega t = I_0 \sin \omega t$, with $I_0 = V_0/R$.

The current and voltage reach their maximum and zero values at the same time: they are **in phase** (phase difference 0).

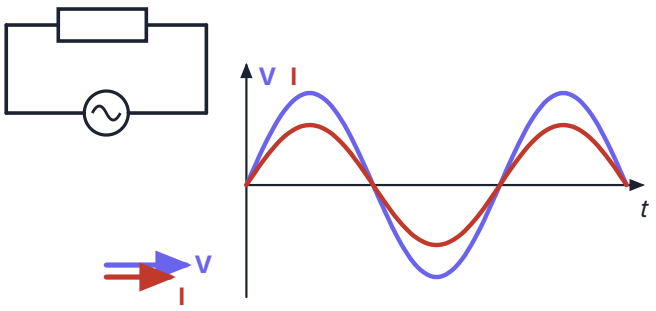


Figure 18.3: a pure resistor: circuit, phasor diagram and graphs.

Power in a pure resistor

$P = IV = I_0 V_0 \sin^2 \omega t = P_0 \sin^2 \omega t$, always positive: the resistor always takes energy and turns it into heat.

Mean of $\sin^2 \omega t = ½$, so:

$$\langle P \rangle = ½ I_0 V_0 = I_{rms} V_{rms} = I_{rms}^2 R = \frac{V_{rms}^2}{R}$$

average power = half the peak power $P_o = I_o V_o$

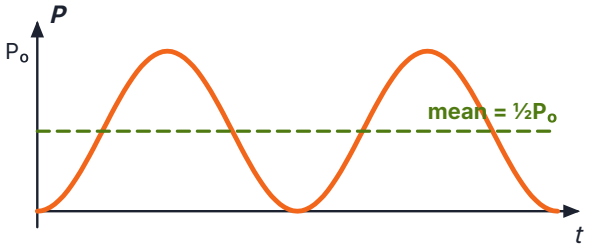


Figure 18.4: power in a resistor varies between 0 and P_o at twice the supply frequency.

Example 18.1.1

A 240 V (r.m.s.), 50 Hz supply is connected to a 60 Ω heater. Find (a) the peak voltage, (b) the r.m.s. and peak currents, (c) the mean power, (d) the peak power.

Solution

(a) $V_0 = \sqrt{2}(240) =$ **339 V**

(b) $I_{\text{rms}} = \dfrac{240}{60} =$ **4.0 A**; $I_0 = \sqrt{2}(4.0) =$ **5.7 A**

(c) $\langle P \rangle = I_{\text{rms}} V_{\text{rms}} = (4.0)(240) =$ **960 W**

(d) $P_0 = I_0 V_0 = 2\langle P \rangle =$ **1920 W**

Example 18.1.2

The current in a resistor is $I = 3.0 \sin 100\pi t$ A. Find (a) the frequency, (b) the r.m.s. current, (c) the current at $t = 2.5$ ms.

Solution

(a) $\omega = 100\pi$, $f = \dfrac{\omega}{2\pi} =$ **50 Hz**

(b) $I_{\text{rms}} = \dfrac{3.0}{\sqrt{2}} =$ **2.1 A**

(c) $I = 3.0 \sin(100\pi \times 0.0025) = 3.0 \sin(\pi/4) =$ **2.1 A**

18.2 Alternating current through an inductor

A **pure (ideal) inductor** has inductance but no resistance.

Deriving the current

The supply voltage is $V = V_0 \sin \omega t$. With no resistance, the applied voltage just balances the back e.m.f.: $V = L\frac{dI}{dt}$.

$$\frac{dI}{dt} = \frac{V_0}{L} \sin \omega t \Rightarrow I = -\frac{V_0}{\omega L} \cos \omega t$$

$I = I_0 \sin(\omega t - \frac{\pi}{2})$, with $I_0 = \frac{V_0}{\omega L}$.

In an inductor the **current lags the voltage by π/2** (a quarter of a cycle): the voltage reaches its peak a quarter-cycle before the current. (Equivalently, V leads I by π/2.)

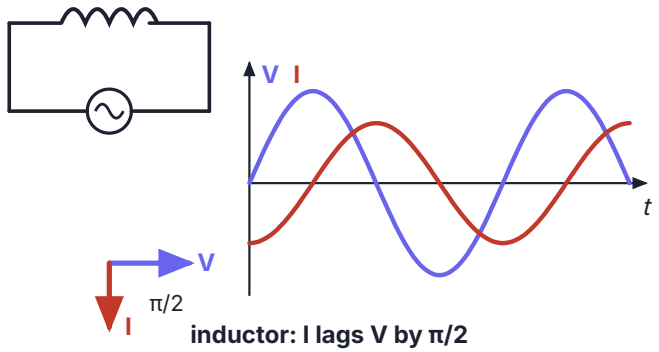


Figure 18.5: a pure inductor: circuit, phasor diagram and graphs.

Inductive reactance, X_L

the **opposition of an inductor to alternating current**: the ratio of the r.m.s. (or peak) voltage across it to the r.m.s. (or peak) current through it.

$$X_L = \frac{V_0}{I_0} = \omega L = 2\pi f L$$

unit: ohm. $X_L \propto f$: an inductor passes d.c. (f = 0) freely but opposes high frequencies.

Power in a pure inductor

$P = IV = I_0(-\cos \omega t) \times V_0 \sin \omega t = -\frac{1}{2} I_0 V_0 \sin 2\omega t$

This is positive for half the time (energy stored in the magnetic field) and negative for the other half (energy returned to the supply). Its mean over a cycle is zero.

$$\langle P \rangle = 0$$

a pure inductor (or capacitor) uses no energy on average

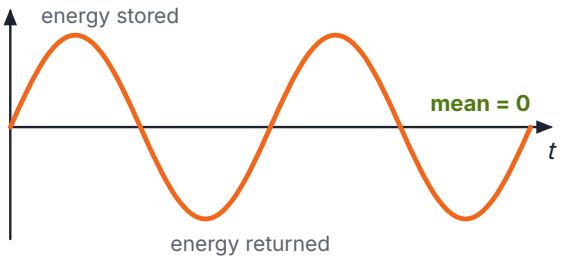


Figure 18.6: power in a pure inductor or capacitor: energy is stored and returned every half-cycle.

Example 18.2.1

A coil of inductance 0.50 H and negligible resistance is connected to a 120 V, 60 Hz supply. Find the r.m.s. current.

Solution

$$X_L = 2\pi f L = 2\pi(60)(0.50) = 188\ \Omega$$

$$I_{rms} = \frac{120}{188} = \mathbf{0.64\ A}$$

Example 18.2.2

A 240 V, 50 Hz supply drives a current of 3.0 A through a pure inductor. Find its inductance.

Solution

$$X_L = \frac{240}{3.0} = 80\ \Omega$$

$$L = \frac{X_L}{2\pi f} = \frac{80}{2\pi(50)} = \mathbf{0.25\ H}$$

18.3 Alternating current through a capacitor

A **pure (ideal) capacitor** has capacitance but no resistance or inductance.

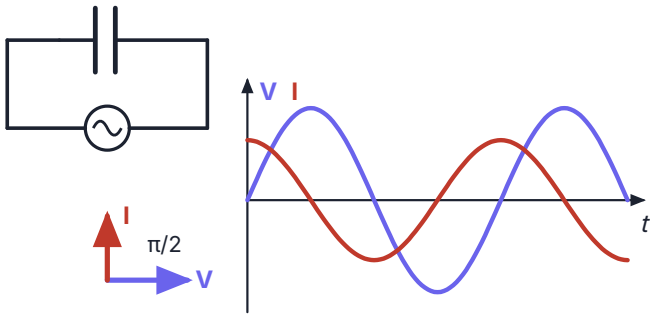
Deriving the current

With $V = V_0 \sin \omega t$ across it, the charge on the capacitor is $Q = CV = CV_0 \sin \omega t$.

$$I = \frac{dQ}{dt} = \omega C V_0 \cos \omega t$$

$$I = I_0 \sin(\omega t + \frac{\pi}{2}), \text{ with } I_0 = \omega C V_0.$$

In a capacitor the **current leads the voltage by π/2**: the current is largest when the capacitor is uncharged (V = 0) and zero when it is fully charged.



capacitor: I leads V by π/2

Figure 18.7: a pure capacitor: circuit, phasor diagram and graphs.

Teacher's note

A memory aid: **CIVIL**. In a **C**apacitor, **I** leads **V**; **V** leads **I** in an inductor (**L**).

Capacitive reactance, X_C

the **opposition of a capacitor to alternating current**: the ratio of the r.m.s. (or peak) voltage across it to the r.m.s. (or peak) current.

$$X_C = \frac{V_0}{I_0} = \frac{1}{\omega C} = \frac{1}{2\pi f C}$$

unit: ohm. $X_C \propto 1/f$: a capacitor blocks d.c. but passes high frequencies easily.

Power in a pure capacitor

$P = IV = I_0 \cos \omega t \times V_0 \sin \omega t = \tfrac{1}{2} I_0 V_0 \sin 2\omega t$.

Energy is stored in the electric field while the capacitor charges and returned while it discharges: $\langle P \rangle = 0$ (Figure 18.6).

Example 18.3.1

An 8.00 μF capacitor is connected to a 150 V (r.m.s.), 60.0 Hz supply. Find the capacitive reactance, the r.m.s. current and the peak current.

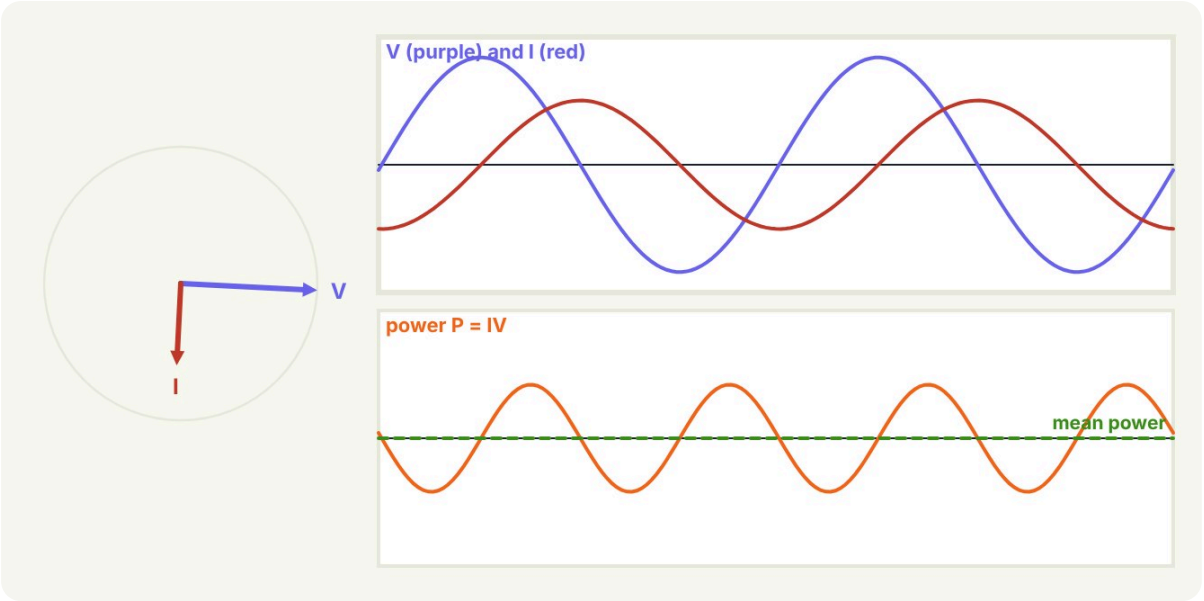
Solution

$$X_C = \frac{1}{2\pi f C} = \frac{1}{2\pi(60.0)(8.00 \times 10^{-6})} = \mathbf{332\ \Omega}$$

$$I_{rms} = \frac{150}{332} = \mathbf{0.452\ A};\ I_0 = \sqrt{2}\,(0.452) = \mathbf{0.639\ A}$$

Simulation R, L or C on an a.c. supply

Choose a resistor, an inductor or a capacitor. The rotating phasors and the graphs show the phase of the current against the voltage; the power curve shows why only the resistor uses energy on average. Change the frequency and watch the reactance.



phase of I relative to V	reactance X_L = 2πfL	peak current (V_o = 10 V)	average power
I lags V by π/2	**100.5 Ω**	**99.5 mA**	**0 W**

This is a still picture. Open PhysiQ › Notes › Semester 2 › Chapter 18 to use the simulation.

Exercise: single elements

1. At what frequency does a 2.0 μF capacitor have a reactance of 400 Ω?

   Answer

   $f = 1/(2\pi X_C C)$ = **199 Hz**

2. The frequency of the supply to a pure inductor is doubled while the r.m.s. voltage stays the same. What happens to the r.m.s. current?

   Answer

   $X_L = 2\pi f L$ **doubles, so the current is halved.**

3. Explain why the average power taken by a pure capacitor is zero.

Answer

In one quarter-cycle the capacitor charges and stores energy taken from the supply; in the next quarter-cycle it discharges and returns the same energy. Over a whole cycle the net energy taken is zero.

18.4 R-C, R-L and R-L-C circuits in series

In a series circuit the **same current** flows through every component, so the current is used as the reference phasor. The voltages across the components have different phases, so they add as **phasors** (vectors), not as numbers.

> **Impedance, Z**
>
> the **total opposition of a circuit to alternating current**: $Z = V_{rms}/I_{rms} = V_0/I_0$. It combines resistance and reactance; unit: ohm.

R-C series circuit



R-C: V lags I by φ **impedance triangle**

Figure 18.8: R-C series: V_R is in phase with I, V_C lags I by π/2.

$$V^2 = V_R^2 + V_C^2 = (IR)^2 + (IX_C)^2$$

$$Z = \frac{V}{I} = \sqrt{R^2+X_C^2}, \quad \tan\varphi = \frac{X_C}{R} = \frac{1}{\omega CR}$$, the current **leads** the voltage by φ.

R-L series circuit



Figure 18.9: R-L series: V_L leads I by π/2.

$V^2 = V_R^2 + V_L^2$

$Z = \sqrt{R^2+X_L^2}$, $\tan\varphi = \frac{X_L}{R} = \frac{\omega L}{R}$, the voltage **leads** the current by φ.



Figure 18.10: how the impedance of R-C and R-L circuits changes with frequency.

R-L-C series circuit

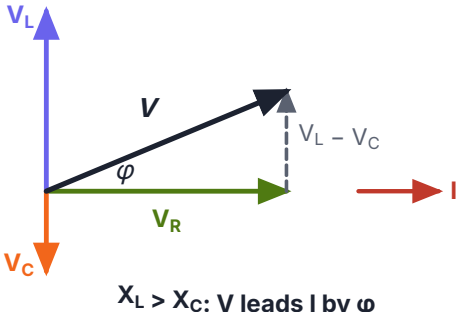


Figure 18.11: R-L-C series: V_L and V_C point in opposite directions, so they partly cancel.

V_L leads I by π/2 and V_C lags I by π/2, so they are in **opposite** directions on the phasor diagram. Their resultant is $(V_L - V_C)$.

$V^2 = V_R^2 + (V_L - V_C)^2 = I^2[R^2 + (X_L - X_C)^2]$

$$Z = \sqrt{R^2 + (X_L - X_C)^2} \qquad \tan\varphi = \frac{X_L - X_C}{R}$$

$X_L > X_C$: V leads I (inductive). $X_L < X_C$: V lags I (capacitive).

Teacher's note

The r.m.s. voltages across the components do **not** add up to the supply voltage. Only the phasor sum does. A voltmeter across the inductor alone can read more than the supply!

Example 18.4.1

A 10 kΩ resistor and a 0.10 μF capacitor are in series with an a.c. supply of angular frequency 1.0×10^4 rad s^{-1}. The r.m.s. voltage across the resistor is 20 V. Find the r.m.s. voltage across the capacitor.

Solution

$$X_C = \frac{1}{\omega C} = \frac{1}{(1.0 \times 10^4)(0.10 \times 10^{-6})} = 1.0\ \text{k}\Omega$$

$$I = \frac{20}{10 \times 10^3} = 2.0\ \text{mA}, \text{ so } V_C = IX_C = \mathbf{2.0\ V}$$

Example 18.4.2

A series R-L-C circuit has $R = 25.0\ \Omega$, $C = 50.0\ \mu\text{F}$ and $L = 0.300$ H, and is driven by a 120 V, 60 Hz supply. Find (a) the impedance, (b) the r.m.s. current, (c) the phase angle between the voltage and the current.

Solution

$$X_L = 2\pi(60)(0.300) = 113.1\ \Omega;\ X_C = \frac{1}{2\pi(60)(50.0 \times 10^{-6})} = 53.1\ \Omega$$

(a) $Z = \sqrt{25.0^2 + (113.1 - 53.1)^2} = \mathbf{65.0\ \Omega}$

(b) $I_{rms} = \frac{120}{65.0} = \mathbf{1.85\ A}$

(c) $\tan\varphi = \frac{60.0}{25.0}$, $\varphi = \mathbf{67.4°}$; $X_L > X_C$, so the voltage leads the current.

Resonance in an R-L-C circuit

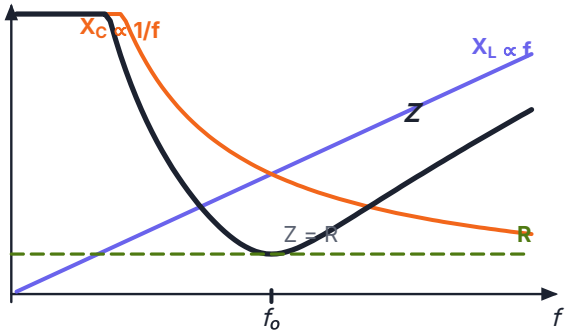


Figure 18.12: X_L, X_C, R and Z against frequency. Z is smallest, and equal to R, at f_o.

- At low frequency X_C is large; at high frequency X_L is large. In between there is a frequency where they are equal.
- When $X_L = X_C$, V_L and V_C cancel exactly: $Z = R$ (its **minimum**), the current is a **maximum** ($I = V/R$) and it is **in phase** with the supply voltage ($\varphi = 0$).

> **Resonance**
> in a series R-L-C circuit occurs when the **frequency of the supply makes the inductive reactance equal to the capacitive reactance**, so the impedance is a minimum (equal to R) and the current is a maximum.

Deriving the resonance frequency

Condition: $X_L = X_C \Rightarrow 2\pi f_0 L = \dfrac{1}{2\pi f_0 C}$

$f_0^2 = \dfrac{1}{4\pi^2 LC}$:

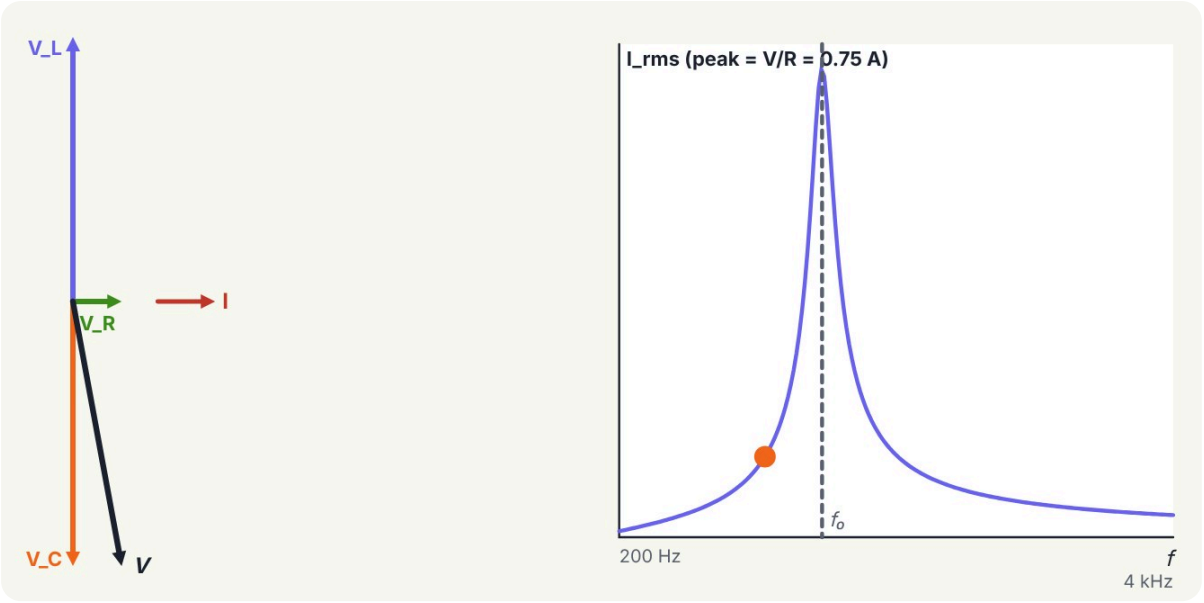
$$f_0 = \frac{1}{2\pi\sqrt{LC}} \qquad \omega_0 = \frac{1}{\sqrt{LC}}$$

independent of R. A smaller R gives a taller, sharper current peak.

- Tuning a radio: the capacitance is changed until f_0 equals the frequency of the wanted station, so that station gives a large current.

Simulation Series R-L-C circuit and resonance

Choose R-C, R-L or R-L-C and change the frequency. The phasor diagram shows V_R, V_L, V_C and the supply V; the graph shows the current against frequency. Find the resonance: V_L and V_C cancel and the current peaks.



impedance Z	current I_rms	phase angle φ	X_L, X_C	V_R, V_L, V_C
1162 Ω	**0.129 A**	**80.1° (V lags I)**	**1508 Ω, 2653 Ω**	**26, 195, 342 V**

$f_o = 1/(2\pi\sqrt{LC})$ **1592 Hz**

This is a still picture. Open PhysiQ › Notes › Semester 2 › Chapter 18 to use the simulation.

Example 18.4.3

A 50 Ω resistor, a 200 mH inductor and a 0.050 μF capacitor are in series with a 150 V (r.m.s.) supply of variable frequency. Find (a) the resonance frequency, (b) the r.m.s. voltage across each component at resonance, (c) the voltage across the inductor and capacitor together, (d) the equation of the supply voltage at resonance.

Solution

(a) $f_0 = \dfrac{1}{2\pi\sqrt{(0.200)(0.050 \times 10^{-6})}}$ = **1.59 kHz** ($\omega_0 = 1.0 \times 10^4$ rad s^{-1})

(b) $I = \dfrac{150}{50}$ = 3.0 A. $X_L = \omega_0 L$ = 2000 Ω = X_C. V_R = **150 V**, $V_L = V_C$ = (3.0)(2000) = **6.0 kV**

(c) V_L and V_C are equal and opposite: **0 V**

(d) $V_0 = \sqrt{2}\,(150)$ = 212 V: V **= 212 sin(1.0 × 10^4 t)** V

Example 18.4.4

A 200 Ω resistor, a 0.75 H inductor and a capacitor C are in series with a 250 V supply. Resonance occurs at 600 Hz. Find (a) X_L and X_C at resonance, (b) C, (c) the impedance at resonance, (d) the current at resonance.

Solution

(a) $X_L = 2\pi(600)(0.75)$ = **2.83 kΩ** = X_C

(b) $C = \dfrac{1}{2\pi f X_C} = \dfrac{1}{2\pi(600)(2827)}$ = **93.8 nF**

(c) $Z = R$ = **200 Ω**

(d) $I = \dfrac{250}{200}$ = **1.25 A**. Phasor diagram: V_R along I, V_L and V_C equal and opposite.

Exercise: series circuits

1. A 40 Ω resistor and a 0.10 H inductor are in series with a 100 V, 50 Hz supply. Find the current and the phase angle.

   Answer

   X_L = 31.4 Ω, Z = 50.9 Ω, I = **1.97 A**, φ = $\tan^{-1}(31.4/40)$ = **38.1° (V leads I)**

2. What capacitance must be put in series with a 25 mH inductor so that the circuit resonates at 1.0 kHz?

   Answer

   $C = 1/(4\pi^2 f_0^2 L)$ = **1.0 μF**

3. Sketch the phasor diagram for a series R-L-C circuit driven below its resonance frequency and state whether the current leads or lags the voltage.

   Answer

   Below f_o, $X_C > X_L$, so V_C is longer than V_L; the resultant V lies below the current phasor. The current leads the voltage (the circuit is capacitive).